

In this game project, I extended the base functionality by adding several features to enhance gameplay and aesthetics. These include:

- 1- Dynamic Camera System: The camera follows the player as they move, creating a scrollable game world. This was achieved using the `translate()` function to adjust the view based on the player's position.
- 2- Collectables: I added collectable items (dollar signs) that the player must gather to complete the game. These items are randomly placed and tracked using an array of objects.
- 3- Enemies: Moving enemies were introduced to increase the challenge. These enemies patrol a fixed range and deduct a life if they collide with the player.
- 4- Flagpole and Game Completion: A flagpole marks the end of the level. The player must collect all collectables before reaching it to win the game.
- 5- Sound Effects: I incorporated sound effects for jumping, collecting items, falling into canyons, and completing the game to make the experience more immersive.

Difficulties:

One of the most challenging aspects was implementing the collision detection system. Ensuring the player interacts correctly with platforms, enemies, and canyons required careful logic and debugging. Additionally, managing the camera system while keeping all game elements properly anchored was tricky.

Skills Learned:

- Object-Oriented Programming (OOP): I used objects and classes to organize code for platforms, enemies, and collectables.
- Collision Detection: I learned how to implement precise collision checks using bounding boxes and distance calculations.
- Game State Management: I managed game states (e.g., game over, game complete) using Boolean flags and conditional logic.
- Debugging: I used console logs and systematic testing to identify and fix bugs.